

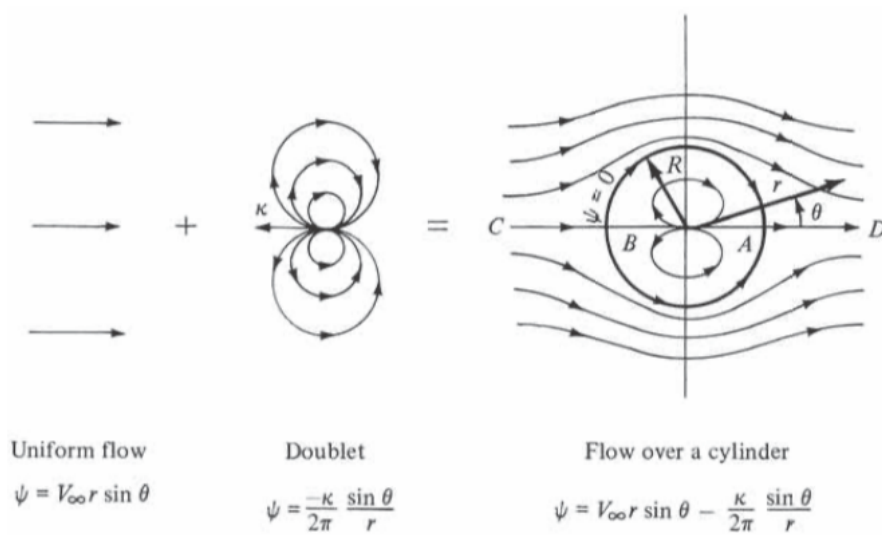


Figure 3.26 Superposition of a uniform flow and a doublet; nonlifting flow over a circular cylinder.

Assumptions

Continuity:
 $\nabla \cdot \vec{v} = 0$

Irrrotational flow
 $\nabla \times \vec{v} = 0$

If $\nabla \times \vec{v} = 0$ then $\exists \phi$ s.t. $\vec{v} = \nabla \phi$ (velocity potential)

Substitute into $\nabla \cdot \vec{v} = 0$
 $\nabla^2 \phi = 0$

or $\frac{\partial^2 \phi}{\partial x_i \partial x_i} = 0$ or $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$



Assumptions + momentum = Bernoulli Equation

$$\Delta^2 \phi = 0$$

what is that?

Laplace Equation

What is it?

$$k = \kappa$$

$$\psi = -\frac{\kappa}{2\pi} \frac{\sin \theta}{r}$$

— Stream function (doublet)

$$\phi = \frac{\kappa}{2\pi} \frac{\cos \theta}{r}$$

— Velocity potential (doublet)

$$\phi = V_{\infty} r \cos \theta$$

— Velocity potential (uniform)

$$\psi = V_{\infty} r \sin \theta$$

— Stream function (uniform)

Adding the

$$\psi = \psi_v + \psi_D = V_{\infty} r \sin \theta - \frac{\kappa}{2\pi} \frac{\sin \theta}{r}$$

$$\psi = (V_{\infty} r \sin \theta) \left(1 - \frac{R^2}{r^2}\right)$$

$$R^2 \equiv \kappa / 2\pi V_{\infty}$$

R — radius of the cylinder

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{1}{r} (V_\infty r \cos \theta) \left(1 - \frac{R^2}{r^2}\right)$$

$$V_r = \left(1 - \frac{R^2}{r^2}\right) V_\infty \cos \theta$$

$$V_\theta = -\frac{\partial \psi}{\partial r} = -\left[(V_\infty r \sin \theta) \frac{2R^2}{r^3} + \left(1 - \frac{R^2}{r^2}\right) (V_\infty \sin \theta)\right]$$

$$V_\theta = -\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin \theta$$

two variable flow

$$\left(1 - \frac{R^2}{r^2}\right) V_\infty \cos \theta = 0$$

$$\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin \theta = 0$$

} $\Rightarrow$ two stagnation points
(R, 0) (R, π)

$$\psi = (V_\infty r \sin \theta) \left(1 - \frac{R^2}{r^2}\right) = 0$$

$$R = \sqrt{\frac{\kappa}{2\pi V_\infty}}$$

goes through both

of them

Symmetrical pressure distribution (no lift) and no drag

$r = R$ - on the surface

$$V_r = 0 \text{ - normal to the surface}$$

$$V_\theta = -2V_\infty \sin \theta \text{ - due to polar coordinates, it's tangential}$$

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2$$

$$V = V_\theta = -2V_\infty \sin \theta$$

$$\frac{V}{V_\infty} = -2 \sin \theta$$

$$\left(\frac{v}{v_0}\right)^2 = 1 - \sin^2 \theta$$

$$1 - \sin^2 \theta$$

$$C_D = \frac{D}{\rho v_0^2 R}$$

$$D = R \int_0^{2\pi} P \cos \theta d\theta$$

$$\int_0^{2\pi} \dots = 2 \cdot \int_0^{\pi}$$